Federal Board HSSC-I Examination
Physics Model Question Paper

Curriculum 2022-2023 (Inclusive Scheme of Studies 2024)

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1	1	1	1	1	1
2	2	2	2	2	2
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Version No.			
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Section - A (Marks 17)

Time Allowed: 25 minutes

Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/ overwriting is not allowed. *Do not use lead pencil.*

Candidate Sign._____ **Invigilator Sign. _____**

Q1. Fill the relevant bubble against each question according to curriculum. Each part carries one mark.

S #	Question	(A)	(B)	(C)	(D)	(A)	(B)	(C)	(D)
i.	Which of the following quantities are dimensionally similar?	Torque and work ✓	Stress and spring constant	Angle and length	Strain and pressure	○	○	○	○
ii.	Maximum work is obtained by the process called:	Isochoric	Isothermal	Isobaric ✓	Adiabatic	○	○	○	○
iii.	Projectile motion is two-dimensional motion under constant acceleration due to	Air resistance	Gravity ✓	Horizontal distance	Velocity	○	○	○	○
iv.	A turtle travel 45cm in 20 minutes, its average velocity will be:	2.25cm/h	90cm/h	135cm/h ✓	15cm/h	○	○	○	○
v.	A stationary wave is formed by waves of frequency 512Hz.The speed of the wave is 65m/s, the distance between two consecutive antinodes is:	0.126m	0.06m ✓	0.03m	0.024	○	○	○	○
vi.	A wire is stretched to double of its original length, its strain is	0.5	1 ✓	zero	2	○	○	○	○
vii.	Cyclotron frequency of a charged particle moving in a magnetic field B is independent of:	Magnetic flux density	Charge on a particle	Velocity of particle ✓	Mass of particle	○	○	○	○
viii.	The best instrument for accurate measurement of emf of a cell is:	Voltmeter	Potentiometer ✓	Ammeter	Ohmmeter	○	○	○	○
ix.	What is the base unit of pressure?	$kgms^{-1}$	$kgm^{-1}s^{-1}$	$kgm^{-1}s^{-2}$ ✓	$kgm^{2}s^{-2}$	○	○	○	○
x.	Theory of relativity was formulated by.	Isaac Newton	Rutherford	Plank	Albert Einstein ✓	○	○	○	○
xi.	The radius at two ends of a pipe is in the ratio of 2:3,then the speed of liquid at the two ends is in the ratio of:	2:3	3:2	4:9	9:4 ✓	○	○	○	○
xii.	Lepton number for muon particle is:	-1	+1 ✓	+1/2	0	○	○	○	○

xiii.	Which of the following moving particle is not deflected by magnetic field?	Neutron ✓	Proton	Electron	Positron	○ ○ ○ ○
xiv.	The force that acts on the body but it does no work is called:	Frictional force	Gravitational force	Centripetal force ✓	Elastic force	○ ○ ○ ○
xv.	The region around a charge in which a test charge can feel an electric force is called	Electric intensity	Electric Potential	Electric Field ✓	Electric Energy	○ ○ ○ ○
xvi.	Moment of inertia of a spinning body about its axis does not depend on the:	Angular velocity of body ✓	Mass of body	Orientation of the axis	Distribution of the mass around axis	○ ○ ○ ○
xvii.	When Fx = 3 N and F = 5 N then Fy =	6N	8N	2N	4N ✓	○ ○ ○ ○

PHYSICS HSSC–I

Inclusive Scheme of Studies 2024 (1200 Marks)

SECTION – A (Marks 17)

Time allowed: 25 Minutes

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Invigilator Sign. _____

Answer Sheet No. _____

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Deleting/overwriting is not allowed.
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Fill the relevant bubble against each question according to curriculum:

	Question	A	B	C	D	A	B	C	D
1.	Which of the following equations is dimensionally correct?	$v_f = v_i + at$	$v_f^2 = v_i^2 + 2at$	$s = v_i t + \frac{1}{2}at$	$F = ma^2$	○	○	○	○
2.	Pascal (Pa) is equal to:	N/m	N/m²	N²/m²	kg/m	○	○	○	○
3.	Time of flight of a projectile depends on:	Horizontal velocity	Mass	Vertical component of velocity	Acceleration only	○	○	○	○
4.	A car accelerates uniformly from rest to $60ms^{-1}$ in 5 seconds. What distance does it cover during this time?	150m	450m	300m	600m	●	○	○	○
5.	Which of the following is a correct way to represent Young's Modulus?	$\frac{F/A}{\Delta L/L}$	$\frac{F.A}{\Delta L}$	$\frac{F}{A.\Delta L}$	$\frac{F.\Delta L}{L.A}$	○	○	○	○
6.	In a steady, incompressible flow, if the cross-sectional area of a pipe decreases to one-third of its original size, what happens to the fluid velocity?	It decreases to one-third	It remains the same	It triples	It becomes nine times larger	○	○	○	○
7.	Direction of centripetal force is always:	Tangential	Perpendicular	Radially inward	Outward	○	○	○	○
8.	If the temperature of heat source is doubled than before, efficiency increases by:	1.5 times	2 times	2.5 times	3 times	○	○	○	○
9.	Antinode in a wave is point of:	Zero displacement	Maximum displacement	Zero velocity	No wave	○	○	○	○
10.	What is the motion of an electron moving at an angle of 45° to a uniform magnetic field?	Linear motion	Circular motion	Parabolic motion	Helical motion	○	○	○	○

	Question	A	B	C	D	A	B	C	D
11.	A potentiometer is used to compare:	Resistance	Current	Potential difference	Power	○	○	○	○
12.	Electric field strength is:	Work done	Energy	Force per unit charge	Charge per force	○	○	○	○
13.	Postulates of special theory of relativity applies to:	Only light	Only motion	All physical laws	Only energy	○	○	○	○
14.	Correct Mass-energy relation is:	E = mv	E = mc²	E = mg	E = gh	○	○	○	○
15.	Proton is made up of:	2 down and 1 up quark	1 down and 2 up quarks	2 down and 2 up quarks	4 down and 2 up quarks	○	○	○	○
16.	If distance from axis increases, moment of inertia:	Decreases	Increases	Remains constant	Becomes zero	○	○	○	○
17.	Correct relation between torque and angular acceleration is:	$\tau = \frac{IR}{\alpha}$	$\tau = \frac{\alpha}{I}$	$\tau = I\alpha$	$\tau = m\alpha$	○	○	○	○

- $S = v_i t + \frac{1}{2}at^2$
- $I = \frac{1}{2}MR^2$
- $F_c = \frac{mv^2}{r}$
- $E = \frac{F}{q}$
- $V = \frac{W}{q}$
- $\eta = 1 - \frac{T_2}{T_1}$
- $I = \frac{2}{5}MR^2$
- $I = mr^2$
- $A_1v_1 = A_2v_2$
- $v_f = v_i + at$
- $E = mc^2$
- $Av = \text{constant}$

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PHYSICS HSSC–I

Inclusive Scheme of Studies 2024 (1200 Marks)

SECTION – A (Marks 17)

Time allowed: 25 Minutes

Candidate Sign. _____

Invigilator Sign. _____

Answer Sheet No. _____

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9/10

Fill the relevant bubble against each question according to curriculum:

	Question	A	B	C	D	A	B	C	D
1	Which equation correctly relates torque (τ), moment of inertia (I), and angular acceleration (α)?	$\tau = I^2\alpha$	$\tau = \frac{I}{\alpha}$	$\tau = I + \alpha$	$\tau = I\alpha$	○	○	○	○
2	The dimensional formula of velocity is:	$[M^0L^0T^0]$	$[LT^{-1}]$ ✓	$[LT^{-2}]$	$[L^2T^{-1}]$	○	○	○	○
3	Joule can be expressed as:	$kg\ m^{-2}\ s^{-2}$	$kg\ m\ s^{-1}$	$kg\ m^2\ s^{-1}$	$kg\ m^2\ s^{-2}$	○	○	○	○
4	Horizontal velocity in projectile motion:	is zero at max height	increases	decreases	remains constant	○	○	○	○
5	A bus travelling at $10ms^{-1}$ accelerates uniformly at $2ms^{-2}$. Calculate the velocity of the bus after 5 s.	$0.02ms^{-1}$	$20ms^{-1}$	$2.0ms^{-1}$	$0.2ms^{-1}$	○	○	○	○
6	Which of the following is a correct way to represent Young's Modulus?	$\frac{F.\Delta L}{L.A}$	$\frac{F/A}{\Delta L/L}$ ✓	$\frac{F.A}{\Delta L}$	$\frac{F}{A.\Delta L}$	○	○	○	○
7	Continuity equation is based on:	Force	Energy conservation	Momentum	Mass conservation	○	○	○	○
8	If velocity of a body moving in circular path doubles, centripetal force becomes:	Four times	Same	Double	Half	○	○	○	○
9	According to the first law of thermodynamics, if 200 Joules of heat is added to a system and 80 Joules of work is done by the system, the internal energy change is:	120J decrease	80J increase	120 J increase	280J increase	○	○	○	○
10	Stationary waves are formed by:	Diffraction	Reflection only	Superposition of waves	Refraction	○	○	○	○

II

	Question	A	B	C	D	A	B	C	D
11.	Path of charged particle in magnetic field is always:	Rondon	Straight	Parabolic	Circular	○	○	○	○
12.	At balance point, current in galvanometer is:	Zero	Maximum	Minimum	Infinite	○	○	○	○
13.	If 800J energy is used to move a charge of 4C between two points in an electric field. Potential difference is:	$5V$	$100V$	$200V$	$5\times0^{-3}V$	○	○	○	○
14.	The rest energy of mass 2.00g is:	$1.80\times10^{18}J$	$180\times10^{12}J$	$1.80\times10^{14}J$	$1.80\times10^{16}J$	○	○	○	○
15.	According to special theory of relativity time dilation means:	Time increases for moving observer	Time decreases for moving observer	Time becomes zero	Time is constant for moving observer	○	○	○	○
16.	Neutrino is:	Magnetic particle	Charged particle	Heavy particle	Neutral particle	○	○	○	○
17.	A solid cylinder with mass M=5kg and radius R=0.3m rotates about its central axis. Calculate the moment of inertia of the cylinder.	$0.23kgm^2$	$5.3kgm^2$	$4.7kgm^2$	$1.5kgm^2$	○	○	○	○

- $v_f = v_i + at$
- $I = \frac{1}{2}MR^2$
- $F_c = \frac{mv^2}{r}$
- $E = \frac{F}{q}$
- $V = \frac{W}{q}$
- $E = mc^2$
- $I = \frac{2}{5}MR^2$
- $I = MR^2$

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PHYSICS HSSC–I
(Curriculum 2022-23) (OLD)
SECTION – A (Marks 17)
Time allowed: 25 Minutes

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Answer Sheet No. _____

Deleting/overwriting is not allowed.
Do not use lead pencil.

Fill the relevant bubble against each question according to curriculum:

	Question	A	B	C	D	A	B	C	D
1.	The dimensions of torque are:	$[M^{-1}L^{3}T^{-2}]$	$[ML^{2}T^{-2}]$	$[M^{2}L^{-1}T^{-2}]$	$[ML^{2}T^{-1}]$	○	○	○	○
2.	If the percentage error in mass is 1% and velocity is 2% then the possible percentage error in momentum is:	3%	4%	5%	8%	○	○	○	○
3.	For which angle the equation $\|A \times B\| = \|A.B\|$ is correct?	30°	45°	60°	76°	○	○	○	○
4.	The F_x component of a force $\vec{F}$ of magnitude 50N making an angle of 60° with X-axis, is:	25 N	5 N	15 N	10 N	○	○	○	○
5.	Two balls of mass 10 kg and 2 gm, collide head-on elastically. If 2 gm ball is stationary and 10 kg ball is moving at 2 m/s before the collision, what is the velocity of 2 gm ball after collision?	0 m/s	4 m/s	3 m/s	6 m/s	○	○	○	○
6.	A projectile has maximum range at:	90°	60°	45°	30°	○	○	○	○
7.	1 radian =	67.3°	60°	5.73°	57.3°	○	○	○	○
8.	If mass of a moving body is doubled but velocity is reduced to one-fourth, then:	K.E increases 2 times	K.E remains the same	K.E decreases 8 times	K.E increases 4 times	○	○	○	○
9.	An ideal fluid has:	Variable velocity	High compressibility	Variable density	Zero viscosity	○	○	○	○
10.	Materials that do not undergo plastic deformation but suddenly break within Elastic region are called:	Amorphous	Ductile	Brittle	Polymers	○	○	○	○

	Question	A	B	C	D	A	B	C	D
11.	In an isochoric process:	Q = 0	Q = ΔW	Q = ΔU	Q = ΔU - ΔW	○	○	○	○
12.	Two tuning forks of frequencies 512 Hz and 516 Hz are sounded together. The number of beats produced per second is	512 Hz	516 Hz	1028 Hz	4 Hz	○	○	○	○
13.	Two charges are experiencing a force of 50 N when placed at 10 cm distance. If charges are placed at 5 cm distance, then force will be:	200 N	75 N	12.5 N	100 N	○	○	○	○
14.	The resistance of a certain length of wire of area 8 mm^2 is 30Ω. If the area is changed to 24 mm^2, then new resistance is:	10 Ω	30 Ω	40 Ω	50 Ω	○	○	○	○
15.	The Time Period of cyclotron is given by:	$T = \frac{qB}{2\pi m}$	$T = \frac{qB}{4\pi m}$	$T = \frac{2\pi m}{qB}$	$T = \frac{qB}{4m}$	○	○	○	○
16.	A passenger passes an electricity pole placed along the side of a road with a speed $\frac{c}{2}$. The length of electricity pole observed by him is:	$l = l_\circ$	$l = \frac{\sqrt{3}}{2} l_\circ$	$l = \frac{2}{\sqrt{3}} l_\circ$	$l = \frac{2}{\sqrt{5}} l_\circ$	○	○	○	○
17.	Proton is made up of:	2 down and 1 up quark	1 down and 2 up quarks	2 down and 2 up quarks	4 down and 2 up quarks	○	○	○	○

- $p = mv$
- $A.B = AB\cos\theta$
- $\overline{A} \times \overline{B} = \hat{n}AB\sin\theta$
- $KE = \frac{1}{2}mv^2$
- $R = \frac{\rho L}{A}$
- $F = \frac{Kq_1q_2}{r^2}$
- For $m_1 >> m_2$; $u_2 = 0$ $v_2 = 2u_1$

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